



Experimental Verification of Laminar Gas Flows and Localized Plasma-Induced Perturbations in a Microcavity Plasma Array

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Abstract

Dielectric barrier discharges (DBDs), and in particular microcavity plasma arrays (MCPAs), represent a promising reactor platform for plasma-catalytic studies, as they allow plasma generation directly at catalytic surfaces, thereby providing direct access to the investigation of plasma–surface interactions. In addition to fundamental plasma parameters such as electric fields and species densities, the gas flow through the reactor plays a crucial role in the process. It directly determines the effective treatment time and volume, while the underlying flow dynamics govern the transport of plasma-generated species such as reactive molecules and atoms. Besides the reactor geometry, the discharge itself may influence the flow through induced forces, potentially generating vortices and enhancing mixing, which could improve conversion efficiency. While complex flow fields may be desirable for maximizing process efficiency, research reactors such as the MCPA benefit from predictable transport conditions when the primary objective is the study of plasma–surface interactions. In this study, the flow dynamics within an MCPA reactor are investigated using particle image velocimetry (PIV) in helium. Spatially and temporally resolved measurements are used to analyze the influence of the plasma discharge on the gas flow. The results show that the discharge can exert a measurable influence on the flow field; however, this effect is spatially confined to a region within approximately 0.5 mm distance from the electrode and is most pronounced during the ignition phase of the discharge. During steady-state operation, the plasma has no significant impact on the overall flow behavior. Under these conditions, the velocity field can be described by a classical laminar Poiseuille profile.

Keywords Dielectric barrier discharge · Particle image velocimetry · Microcavity plasma array · Plasma catalysis

Introduction

Dielectric barrier discharges (DBDs) have attracted considerable attention in recent years due to their strong potential for converting volatile organic compounds (VOCs) using non-thermal plasmas [1–3]. These plasmas enable the activation of chemical reactions at near-ambient gas temperatures, offering a promising pathway for energy-efficient pollutant abatement and chemical processing. In particular, the combination of DBDs with catalytic materials, commonly referred to as plasma catalysis, has been shown to enhance conversion efficiency, tailor product selectivity, and/or induce synergistic effects between plasma-generated species and surface-mediated reactions [4–7].

To enable a deeper understanding of plasma–surface interactions, a specialized DBD configuration, the microcavity plasma array (MCPA) reactor, was introduced [8, 9]. This reactor contains thousands of microcavities with diameters and depths in the micrometer range, in which individual microdischarges are confined. The geometric confinement leads to a high surface-to-volume ratio and consequently to a strong interaction between the plasma and the surrounding surfaces. The modular design of the reactor allows different (catalytic) materials to be easily exchanged and characterized before and after plasma operation. In addition, the reactor provides optical access, enabling the application of spectroscopic and laser-based diagnostic techniques. These unique features make the MCPA reactor a powerful platform for fundamental studies in plasma catalysis. Previous investigations using this reactor concept have addressed parameters such as electric field distributions [9, 10], surface charge accumulation [11], discharge dynamics [12], and the production and transport of reactive species [13–15].

Molecular dissociation within the microcavities is highly efficient, confirming that the confined microdischarges provide a chemically very active environment [13, 15, 16]. Nevertheless, the overall reactor behavior is not determined by the plasma chemistry alone, but is also significantly influenced by transport processes that control how reactant molecules reach and leave the active regions. In the current reactor configuration, the main gas stream flows predominantly along the electrode surface, while the exchange between the bulk gas and the plasma-filled cavities occurs mainly via diffusion [14, 16]. Consequently, only a limited fraction of the total gas flow effectively interacts with the active plasma volume. Previous studies have consistently assumed a laminar gas flow, which has been able to successfully describe the observed behavior and experimental results [14, 16], although this has so far only been inferred indirectly from experiments and has not yet been directly investigated. This well-defined laminar flow is advantageous for the MCPA as a research platform, as it allows for a simplified and predictable description of transport processes, thereby enabling a more detailed investigation of fundamental plasma–surface interactions.

However, this result is somewhat surprising, as plasma discharges are well known to influence gas dynamics. Plasma-induced electrohydrodynamic (EHD) forces, for example, can modify local flow fields and generate complex flow structures [17, 18]. Such effects have been reported in comparable DBD systems (e.g., actuator studies), where in some cases the formation of vortices has been explicitly exploited to enhance conversion efficiency through improved mixing, as demonstrated by Bøddecker et al. [19]. Similar discharge–flow interactions, including vortex formation in helium-driven DBD plasma jets impinging on liquids, have been reported and can significantly affect reactive species transport, thereby influencing conversion pathways in plasma-driven processes [20–22].

A powerful technique to experimentally investigate gas flow dynamics is particle imaging velocimetry (PIV) [23]. In PIV, tracer particles are introduced into the gas flow and illuminated by a high-intensity light source, such as a laser sheet. The scattered light from the particles is recorded by a high-speed camera. By correlating two images acquired with a short and known time delay, the spatial particle image displacement and thus the local velocity field can be determined [23].

The aim of this work is to systematically investigate the gas flow dynamics within the MCPA system. In particular, direct PIV measurements in close proximity to the discharge are conducted to resolve the influence of the plasma on the local flow field. This approach allows for a direct assessment of whether the gas flow can be accurately described by a laminar profile, which would further underline the value of the MCPA as a research platform for fundamental plasma-catalytic studies and justify the continued use of simplified transport models for species within the reactor. At the same time, the study explores the extent to which plasma-induced flow effects may be present and whether such effects could be deliberately exploited in future MCPA designs, for instance to enhance conversion efficiency.

Experimental Setup

Plasma Source

The plasma source used in this work is a dielectric barrier discharge (DBD), specifically a microcavity plasma array (MCPA) and is illustrated in Fig. 1. A neodymium permanent magnet (50 mm × 15 mm × 5 mm), mounted in a polyetheretherketone (PEEK) carrier, functions as the grounded electrode. It is separated from a nickel grid connected to the high-voltage supply by a 40 µm thick zirconium dioxide dielectric layer (88 mm × 27 mm, relative permittivity $\epsilon_r \approx 27$), forming the dielectric barrier. The zirconium dioxide dielectric was selected due to its high relative permittivity, which enhances the effective capacitance of the discharge and promotes a more homogeneous ignition across the microcavity array. In

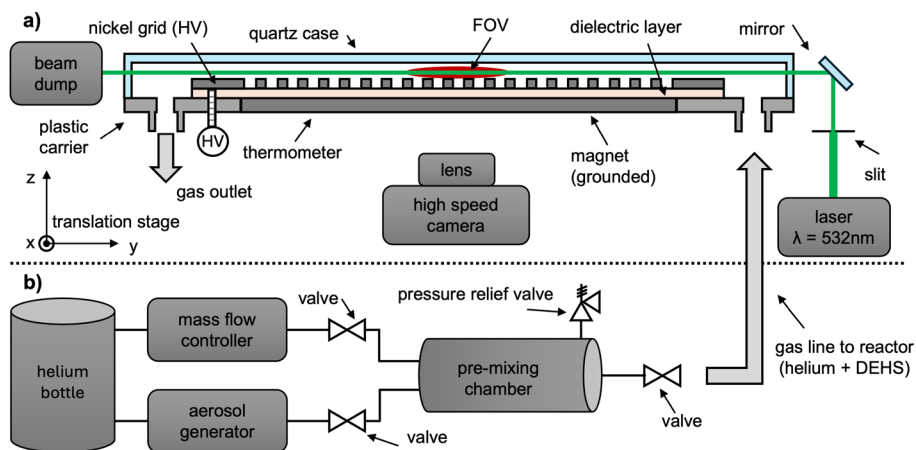


Fig. 1 Schematic sketch of (a) the microcavity plasma array reactor, including the optical PIV setup, and (b) the gas supply system for generating a helium-DEHS mixture

addition, the dielectric barrier limits the discharge current via surface charge accumulation, preventing transition to arc discharges. The relatively small thickness of 40 μm further facilitates efficient coupling of the applied electric field into the microcavities. The nickel grid is fabricated by laser cutting a 50 μm thick nickel foil and contains several thousands of microcavities with diameters of 200 μm and a spacing of 150 μm . The stack consisting of a magnet, a dielectric foil, and a nickel grid is assembled via magnetic attraction. This design provides a modular configuration, enabling rapid exchange of dielectric materials or electrode grids as well as straightforward replacement of damaged components. The applied magnetic force is sufficiently strong to ensure mechanical stability and has a negligible influence on the plasma, since the electron collision frequency greatly exceeds the corresponding gyrofrequency in the magnetic field [8]. The complete arrangement is enclosed by a quartz case with a width of 2 cm and a clear height of 4 mm. This enclosure provides optical access for diagnostic purposes while simultaneously defining controlled boundary conditions for the gas flow across the cavity array. The resulting confined geometry stabilizes the flow profile above the microcavities and determines the residence time and transport of reactive species, which are relevant for plasma–surface interactions and overall process performance.

The reactor is operated under atmospheric conditions using helium as the working gas with flow rates of up to 2 slm. The driving voltage waveform is generated by a function generator (Tektronix AFG 3021B) and amplified using a high-voltage amplifier (Trek Model PZD700A M/S). In this study, a bipolar ramp voltage with a peak-to-peak voltage of 1200 V at a frequency of 15 kHz is applied. The Cartesian coordinate system commonly adopted for the reactor geometry is indicated in Fig. 1. Additional information on the reactor concept and characterization is provided in refs [8, 9].

PIV Diagnostics

For PIV measurements, it is necessary to seed the helium gas flow with tracer particles, most suitably aerosols. In this work, di-ethyl-hexyl-sebacate (DEHS) is used as seeding material, forming fine droplets in the order of 1 μm . DEHS is characterized by well-controllable and reproducible particle sizes, favorable optical properties, chemical inertness toward typical plasma species, and a low tendency for electrical charging. Furthermore, DEHS does not tend to condense on surfaces, which is a major advantage in micro cavities as it prevents them from clogging up [23–25].

The aerosol is generated using an aerosol generator (LaVision GmbH, Germany), which is typically operated with compressed air. However, since a controlled and pure helium gas flow is required in this study, a dedicated mixing system was developed (see Fig. 1b). The aerosol generator is directly supplied with helium, producing a helium–DEHS aerosol mixture. This mixture is directed via a valve into a pre-mixing chamber. The outlet of the chamber is initially closed, allowing a slight overpressure to build up, which can be regulated by a pressure relief valve. Once the chamber is filled, the connection to the aerosol generator is interrupted by closing a valve. For the PIV measurement, a desired helium flow rate (100 sccm – 2 slm) is set using a mass flow controller (MKS MFC GE50A). When the valve toward the reactor is opened, a controlled helium flow passes through the pre-mixing chamber and entrains the generated aerosol into the reactor. For the operating conditions used in this work, the amount of seeding stored in the pre-mixing chamber is sufficient to

sustain PIV measurements for up to five minutes, which is more than adequate compared to typical measurement times of only a few seconds.

For PIV diagnostics, a laser (Coherent Verdi V6, $\lambda = 532$ nm) with an output power of 6 W is used. The laser beam is shaped using a slit aperture and subsequently guided through the reactor by means of a mirror before being terminated in a beam dump. The slit is adjusted such that the laser sheet has a width of approximately 0.5 mm in the x -direction and a height of approximately 2.5 mm in the z -direction. The laser is operated with active cooling (Coherent ThermoTek T255P).

A high speed camera (Phantom Veo 410L) is used for image acquisition and is equipped with a long-distance microscope (KX InfiniMax, MX-6), enabling high resolution imaging of the seeded flow. The camera is positioned to view the region between the metal grid and the quartz cover from the side, and it is aligned at a right angle to the laser sheet. Based on the camera's field of view (FOV), the position of the laser sheet, and the seeding distribution, an elliptically shaped region of interest is defined for the PIV evaluation. The spatial resolution is approximately $5.76 \mu\text{m pixel}^{-1}$ and is calibrated using a millimeter scale placed at the position of the laser sheet plane, when the laser was turned off. The effective measurement region is thus restricted to the illuminated section of the flow, with its extent defined by the intersection of the laser sheet and the camera field of view. The final vector field resolution depends on the chosen interrogation window size and overlap during PIV processing, while the minimum feasible window size is mainly constrained by the seeding density and the frame rate.

In Particle Image Velocimetry, the displacement of particle patterns between two consecutive images is determined by applying cross-correlation on small interrogation windows to obtain the local velocity field. The temporal separation between the image pairs was set to $\Delta t = 118.8 \mu\text{s}$ (8420 fps) for a flow rate of 2000 sccm and $\Delta t = 188.79 \mu\text{s}$ (5297 fps) for all lower gas flow rates. Operating the camera at the high frame rate of 8420 fps leads to a reduction of the image resolution from 1280×800 pixel to 1152×552 pixel. Under discharge operation, the resulting particle image displacements satisfied in this way for all flow rates the so-called one-quarter criterion, i.e., the particle displacement between consecutive frames remained approximately four times smaller than the corresponding interrogation window size. Fulfilling this condition reduces in-plane loss of particle pairs and thereby improves the robustness and accuracy of the cross-correlation analysis [26]. The PIV evaluation was conducted using the software DaVis (Version 10, LaVision GmbH, Germany). Spatial calibration was performed at the beginning of each measurement series by placing a millimeter scale in the plane of the laser light sheet, enabling the determination of the calibration factor.

The plasma emission was monitored using a photomultiplier (Hamamatsu R1527, spectral response: 185 nm - 680 nm), operated with a power supply (Hamamatsu C3830) and read out using an oscilloscope (Teledyne WavePro 404HD, 4 GHz bandwidth). The signal was transmitted via a fiber optic cable (Ocean Optics QP600-2-UV-BX).

Since the reactor temperature may influence the gas dynamics in addition to the plasma dynamics, the reactor temperature is continuously monitored. For this purpose, the temperature at the grounded electrode is measured using an optical fiber based fluoroptic thermometer (Luxtron I652).

Measurement Uncertainty

Prior to the PIV evaluation, the recorded images were preprocessed in order to enhance the particle image contrast and to improve the separation of particle images from the background. For this purpose, a sliding minimum subtraction with a kernel size of 3 pixel was applied in the software, effectively subtracting a locally determined constant intensity value from each pixel, followed by multiplication with a scaling factor to increase image contrast. For the evaluation of the PIV measurements, three successive interrogation window sizes of 128×128 pixel, 64×64 pixel, and 32×32 pixel were applied, using window overlaps of 75 % and 50 %, respectively. The minimum interrogation window dimension was selected to ensure that each interrogation region contained at least ten particle images [26]. The optical configuration of the PIV system provides a spatial scaling on the order of $5 \mu\text{m pixel}^{-1}$. Based on the smallest interrogation window size, this corresponds to a vector spacing (spacing between calculated neighboring velocity vectors) of approximately 0.16 mm. After performing the cross-correlation analysis, the resulting velocity fields were subjected to a semi-automated validation procedure in DaVis. This procedure applies an anisotropic denoising filter to the vector field and the interpolation of the outliers as also described in [27, 28]. The accuracy estimation for PIV results is rather complex due to several error contributions originating from particle image size, image noise, and in-plane as well as out-of-plane particle losses. Here, the uncertainty in the determined particle displacements is assessed based on the findings reported in [29]. For a typical noise level (signal-to-noise ratio of the true PIV correlation peak to side peaks) of 0.015 and a particle loss (loss of tracer particles between successive images, e.g. due to out-of-plane motion) of 15 % between consecutive images, the resulting average uncertainty in the computed displacement is approximately ± 0.2 pixel. This corresponds to a flow velocity uncertainty of roughly $\pm 5 \text{ mm s}^{-1}$. Temporal and spatial averaging significantly reduces the statistical uncertainty of the mean velocity. For typical sample sizes of $N \approx 1000$, the uncertainty is reduced by a factor of $N^{-1/2} \approx 0.03$, enabling the detection of small flow components.

Results

Temperature Development Within the Reactor

The reactor was characterized under conditions as close as possible to those encountered during plasma-catalytic operation, in terms of applied voltage, frequency, and gas flow rates. A key parameter in this context is the reactor temperature. Although gas flows are typically introduced at room temperature, continuous discharge operation leads to a gradual heating of the gas and the reactor. Consequently, the following PIV measurements were performed under steady-state thermal conditions representative of typical operating regimes. For this purpose, the temperature at the grounded electrode (magnet) was continuously monitored, as this location allows reliable and non-intrusive measurements without interfering with the discharge or gas flow. While temperature gradients across the reactor assembly are expected due to the localized heating [13] in the plasma region and cooling by the gas flow, the grounded electrode is in close proximity to the discharge. It is therefore assumed that once

a stable temperature is reached at this position, the overall system has attained thermal equilibrium.

Figure 2 shows the temporal evolution of the reactor temperature after ignition of the discharge for different gas flow rates. In all cases, the temperature increased from room temperature over a period of approximately 25 minutes before reaching a steady-state value of about 55 °C. During the initial heating phase, the temperature rise was nearly identical for all investigated flow rates, which can be attributed to the dominant heating by the discharge, while convective cooling by the gas flow remains weak. This is due to the relatively low flow velocities resulting from the large cross-sectional area of the reactor. In addition, the parabolic velocity profile leads to particularly low gas velocities near the surface where the temperature is measured, further reducing the influence of the flow on local heat transfer. Consequently, no significant influence of the adjusted gas flow rate was observed, except for the highest flow rate of 2000 sccm. At 2000 sccm, the temperature curve leveled off slightly earlier and reached a lower steady-state temperature of approximately 52 °C, indicating a noticeable cooling effect of the increased gas flow on the reactor.

Based on these observations, a defined measurement procedure was established. The reactor was first operated until the steady-state temperature was reached, ensuring thermal equilibrium. Subsequently, the reactor was switched off for a maximum of 30 seconds to introduce the seeding particles and prepare the experiments. The downtime was minimized in all cases to allow the measurements to start as close as possible to the steady-state temperature. Overall, temperature deviations of less than 5 K relative to the respective steady-state equilibrium temperature were observed throughout the experimental campaign.

Impact of Plasma on Flow Behavior

In this section, the impact of the plasma on the flow field is investigated using PIV. According to the procedure described above, the seeding particles were introduced prior to each measurement.

Figure 3a shows a reference measurement at a helium flow rate of 250 sccm without plasma ignition. The color scale represents the local velocity magnitude c , with streamlines overlaid to illustrate the flow field. The elliptical structure represents the evaluable region within the gas channel. This region results from the overlap of the optical imaging onto the high-speed camera sensor, the laser sheet profile, and the spatial distribution of the seeding

Fig. 2 Temperature development of the grounded electrode (magnet) as a function of the discharge operating time and the applied gas flow rate. While most gas flow rates show similar dynamics, at 2000 sccm a gas cooling effect is observed

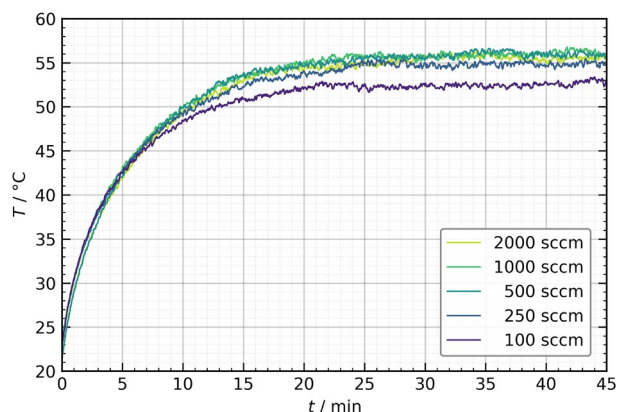
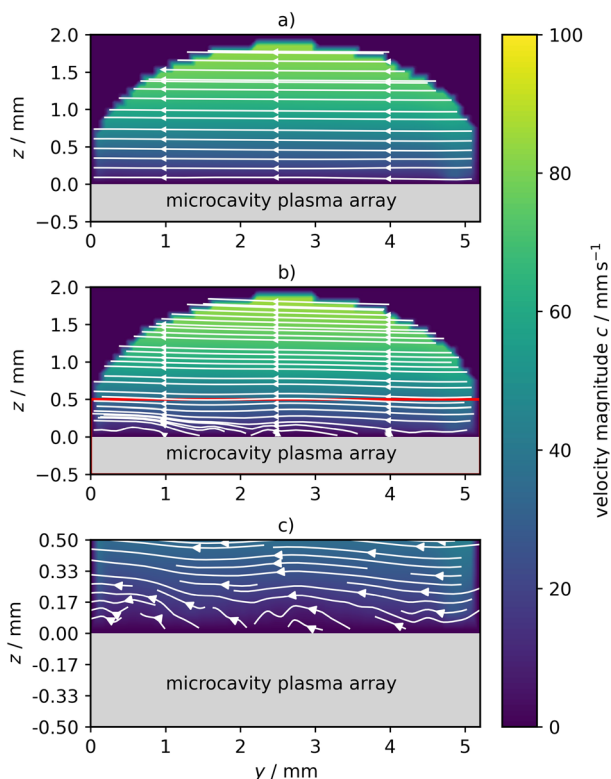


Fig. 3 Streamline representation of the time-averaged flow field for (a) the deactivated array and (b) the operating array for an applied gas flow rate of 250 sccm. (c) shows a digitally magnified view (spatially deformed) of the array surface region while the array is operating, highlighting the near-surface flow structures. The flow fields are averaged over 1000 images at a camera recording frequency of 5297 Hz. For flow calculation close to the array surface, the array was at room temperature in this special case, as thermal convection would induce buoyancy of the seeding particles and prevent meaningful evaluation in this zone



particles. For clarity, the position of the MCPA is indicated in the figure. In order to capture as much information as possible close to the surface, the camera and laser were positioned so that the vertical center of the ellipse was close to the electrode. As a result, however, only 2 mm can be resolved in the z -direction in this configuration.

Overall, a highly homogeneous laminar flow through the channel is observed. Within the accessible measurement region, the velocity decreases toward the electrode, consistent with a parabolic profile. The presence of a velocity maximum near the channel center is expected but not directly resolved in this measurement and is examined in more detail in Sect. "Flow Velocity Profiles as a Function of Temperature". This result could be expected based on the chosen operating parameters, as the Reynolds number for the given conditions (reactor geometry, helium as carrier gas, and flow rates between 250 and 2000 sccm) remains below $Re \leq 30$, indicating laminar flow. Achieving turbulent conditions within the reactor would require significantly higher flow rates on the order of approximately 200 slm for helium, which is well beyond the design limits of the present setup. Even when considering gases with higher density and/or lower viscosity, such as CO_2 in plasma-catalytic applications, flow rates exceeding 10 slm would still be necessary to induce turbulence. In addition, these gases are typically diluted in helium, effectively increasing the total required flow rates even further. Therefore, under plasma-off conditions, the reactor can be assumed to operate consistently in the laminar flow regime. The results are in very good agreement with findings from previous studies [14, 16], in which the transport of reactive species was successfully described based on a laminar gas flow profile.

This flow pattern changes once the discharge is ignited (see Fig. 3b). Near the lower boundary, in the vicinity of the discharge region, the streamlines are no longer strictly parallel but exhibit small perturbations. It appears that the discharge has an influence on the flow dynamics at least in the area up to 0.5 mm away from the electrode, for example, through induced forces or increased temperature. A digitally magnified view of the area is shown in Fig. 3c). These effects will be analyzed in greater detail below. In addition, the flow velocity increases, with the maximum velocity in the channel center rising from approximately 80 mm s^{-1} to about 90 mm s^{-1} .

Despite these local disturbances, the overall flow through the reactor can still be considered as homogeneous and predominantly laminar, even under active plasma conditions. However, the specific influence of the plasma on the flow field will be examined in more detail in the following sections.

Flow Velocity Profiles as a Function of Temperature

One of the most significant differences in the flow field between operation with and without plasma is the increase in flow velocity upon ignition of the discharge. To investigate this effect in more detail, the velocity profile along the z -direction of the channel can be evaluated (see Fig. 4). Compared to the measurement shown previously, the laser and camera were now adjusted so that the largest area possible of the flow channel ($\approx 2.5 \text{ mm}$) could be resolved. Nevertheless, the setup does not cover the entire channel height (4 mm). While this would in principle be possible, expanding the laser sheet would reduce the available intensity and thus the temporal resolution. In addition, increased stray light near the quartz boundaries limits measurement quality, such that only selected regions of the channel are investigated. Furthermore, the seeding particles are not uniformly distributed inside the channel, leading to a higher particle density in the center compared to the edges. This non-uniformity is most likely not caused by particle sticking or condensation at the walls, but rather by flow- and temperature-induced transport effects, including discharge-driven motion and thermally induced buoyancy. If the local particle concentration decreases below a certain threshold, reliable evaluation is no longer possible in those regions.

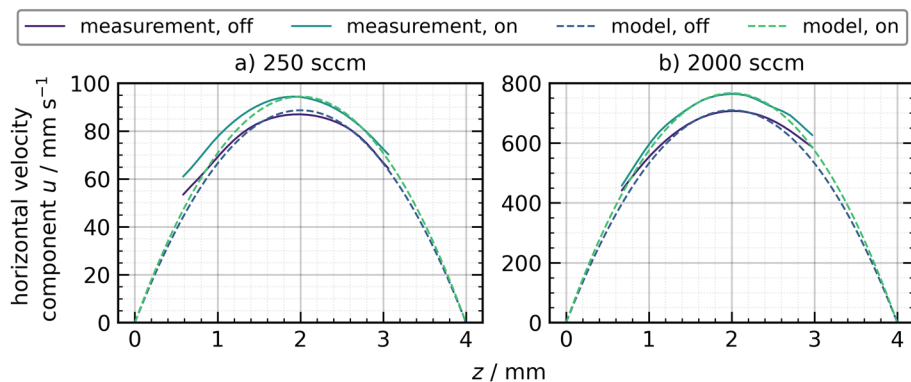


Fig. 4 Vertical velocity component u for a total gas flow of (a) 250 sccm and (b) 2000 sccm as a function of the vertical position z for both plasma-on ("on") and plasma-off ("off") conditions. A basic one-dimensional steady laminar flow model can explain the profiles by an increase in temperature, but shows slight deviations at the edges

To describe the observed effects, a simplified flow model was employed. The flow inside the channel is assumed to be steady, laminar, and one-dimensional. Under these assumptions, the stationary Navier-Stokes equation for pressure-driven channel flow reduces to

$$-\frac{dp}{dy} + \mu \frac{d^2 u}{dz^2} = 0. \quad (1)$$

Here, u denotes the axial flow velocity, p the pressure, and μ the dynamic viscosity. Solving this equation with no-slip boundary conditions at the channel walls yields the classical parabolic Poiseuille velocity profile

$$u(z) = \frac{1}{2\mu} \left(-\frac{dp}{dy} \right) z(L_z - z), \quad (2)$$

where L_z denotes the channel height, which is $L_z = 4$ mm in the present reactor geometry. In the experiment, the pressure gradient is not imposed directly but is determined by the mass flow rate set by the MFC. The mass flow rate $\dot{m}$ can be obtained by integrating the velocity profile across the channel cross section:

$$\dot{m} = \rho L_x \int_0^{L_z} \frac{1}{2\mu} \left(-\frac{dp}{dy} \right) z(L_z - z) dz. \quad (3)$$

Here, L_x represents the channel width and ρ is the mass density. Since the velocity profile is integrated over the entire cross-sectional area, the mass flow depends on both geometric dimensions of the channel. Rearranging the expression yields the corresponding pressure gradient

$$\frac{dp}{dy} = - \frac{2\mu \dot{m}}{L_x \int_0^{L_z} \rho(z) z(L_z - z) dz}. \quad (4)$$

The local gas density is calculated using the ideal gas law

$$\rho(z) = \frac{p_0}{R_{\text{He}} T(z)}, \quad (5)$$

where p_0 is the operating pressure, R_{He} the specific gas constant of helium, and $T(z)$ the gas temperature depending on the position. Assuming a constant gas temperature within the channel, the density becomes uniform and an analytical solution for the velocity profile can be obtained. In this case, inserting the corresponding pressure gradient into the Poiseuille solution yields

$$u(z) = \frac{6\dot{m}}{\rho L_x L_z^3} z(L_z - z). \quad (6)$$

Consequently, the model depends only on the reactor geometry, the imposed mass flow rate, and the gas temperature. The analysis assumes steady and laminar flow, as well as constant temperature and viscosity. Plasma-related effects on the flow are neglected. The model is therefore intended solely to illustrate the fundamental behavior of the system. A more rigorous description of the flow dynamics would require a comprehensive multiphysics model that could also account for location-dependent temperatures or viscosities and buoyancy effects, which is beyond the scope of the present work.

The results for two different flow rates (250 and 2000 sccm) are shown in Fig. 4. Both the measurements and the model exhibit the expected parabolic velocity profile across the channel cross-section. Overall, the model reproduces the absolute velocities very well, particularly in the center of the channel.

Figure 4 also illustrates the difference between plasma-off and plasma-on conditions. For these measurements, the reactor was preheated by operating a plasma until the equilibrium temperature was reached so that the reactor temperature (measured at the grounded electrode) was identical in both cases. As already observed in the previous section, the gas velocity increases when the plasma is switched on, resulting in an upward shift of the entire velocity profile. In the model, this effect can be reproduced by assuming a gas temperature of 55 °C instead of 21 °C.

The assumed gas temperature within the channel is consistent with previous observations. On the one hand, it is known that the entire reactor heats up to temperatures above 50 °C during operation and can therefore transfer heat to the gas flow. On the other hand, measurements indicate temperatures of 200–500 °C within the discharge region [13, 15]. Although the discharge volume is significantly smaller than the channel volume, a temperature increase of about 30 K due to the discharge is therefore realistic. This also implies that the thermal effect caused by the discharge plays a dominant role, while additional plasma effects, such as induced electrohydrodynamic forces, are negligible in this case.

Due to measurement limitations, velocities could not be measured directly at the surfaces. Nevertheless, it can be observed that the velocity profile deviates slightly from the parabolic shape near the electrode ($z = 0$ mm) and is somewhat higher than predicted by the model. This may indicate a small temperature gradient from the heated electrode region to the quartz case. In addition, potential effects arising from surface interactions or from the discharge itself were neglected in the model, so no quantitative statement can be made about them. In the case of 2000 sccm, however, deviations are also observed towards the quartz case ($z = 4$ mm). Here, for example, surface interactions with the quartz window may also play a role. Despite these limitations, the results suggest that the temperature increase caused by the discharge dominates the observed flow behavior, while additional plasma-induced effects, if present, are likely confined to regions very close to the surface.

Dependence on Flow Velocity

It has been shown that the operating temperature can influence the absolute flow velocity, whereas the overall shape of the velocity profile remains largely unaffected by the presence of the discharge. In the following, the influence of the total gas flow rate on the flow dynamics is investigated in more detail.

A suitable quantity to characterize potential changes in the flow structure is the vorticity $\vec{\omega}$, which provides a measure of local rotational motion within the flow field. By analyzing

the vorticity distribution for different total gas flow rates, possible deviations from purely laminar behavior can be systematically assessed.

Here, we calculate the vorticity $\vec{\omega}$ in 2D according to [30] as

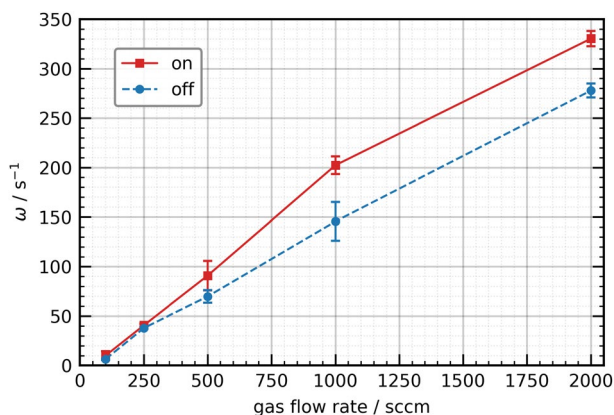
$$\vec{\omega} = \vec{\nabla} \times \vec{c} = \left(\frac{\partial v}{\partial y} - \frac{\partial u}{\partial z} \right) \vec{e}_x. \quad (7)$$

Within the scope of this study, the velocity vector is defined as $\vec{c} = u \vec{e}_y + v \vec{e}_z$, with y and z denoting the horizontal and vertical Cartesian directions, respectively. The unit vector $\vec{e}_x$ points normal to the yz -plane. The information on rotational direction is not of interest in this study, and only the magnitude of the quantity is discussed in the following.

Figure 5 shows the calculated vorticity as a function of the total gas flow rate for both plasma-on and plasma-off conditions. The vorticity increases almost linearly with increasing flow rate in the plasma-off case. This linear behavior is expected, since the vorticity is directly linked to the flow velocity, which under laminar conditions scales proportionally with the imposed gas flow rate. In contrast, for the plasma-on case a reduced slope is observed for flow rates above 1000 sccm. This deviation from linearity is consistent with the earlier observation of a flow-rate dependent cooling effect in the temperature measurements, where higher gas flows lead to more efficient convective heat removal from the reactor. A lower gas temperature would in turn reduce the gas expansion induced by the discharge and thus decrease the effective flow enhancement, which directly affects the measured vorticity. However, this interpretation cannot be fully confirmed based on the present dataset and would require additional measurements at intermediate and higher flow rates to resolve the underlying transition behavior.

The absolute values are increased by up to 50 s^{-1} in the plasma-on case. This increase can primarily be attributed to the higher flow velocity previously observed under plasma operation, which mainly results from the elevated gas temperature. Nevertheless, an additional plasma-specific contribution cannot be excluded. In particular, possible electrohydrodynamic (EHD) forces would act predominantly perpendicular to the main flow direction and could therefore further enhance the local vorticity. This is consistent with the observed deviations from laminar behavior near the surface. However, it is also apparent that such effects are likely to be small and remain within the uncertainty of the present measurements.

Fig. 5 Absolute value of the vorticity ω as a function of the gas flow rate for both plasma-on and plasma-off conditions. The displayed values are spatially averaged over a rectangular region extending across the full width of the measurement area, approximately 1 mm in height and located about 0.25 mm above the surface (to avoid measuring in an area where aerosol is vanishing due to thermal convection)



In principle, increased vorticity could facilitate a faster transport of reactive species out of the discharge cavities and enhance mixing within the reactor. However, the measurements indicate that the vorticity cannot be significantly enhanced by plasma operation and, if present, is confined to regions very close to the surface. For future reactor improvements, the design could be modified to promote higher flow velocities, for example by reducing dead volume, or to intentionally disturb the flow through geometrical features in order to generate additional vortical structures and thereby improve gas mixing.

Temporal Development of Plasma-Induced Flows

The results obtained so far indicate that the plasma, e.g. through induced forces, exerts only a minor influence on the overall flow dynamics within the reactor. To investigate this aspect in more detail, time-resolved scans over a total duration of 2000 ms were performed. Within this time window, the first ignition of the discharge occurs, enabling a direct comparison of the flow conditions immediately before and after gas breakdown.

As shown previously, the flow inside the reactor is dominated by velocity components in the horizontal y -direction. Vertical velocity components were only observed to a limited extent, mainly associated with increased vorticity at higher gas flow rates or in the vicinity of the active (powered) electrode, as shown in Fig. 3. For this reason, the vertical velocity component v represents a particularly sensitive indicator for detecting plasma-induced effects on the flow field.

Figure 6 presents the temporal evolution of the vertical velocity component for different gas flow rates. As expected, higher gas flow rates exhibit larger vertical velocity components from the outset, consistent with the increased vorticity discussed above. During the initial seconds of the measurement, no discharge is present and the flow is solely governed by the imposed gas conditions.

Upon first ignition of the discharge within the measurement window, a distinct influence on the gas dynamics becomes evident. The vertical velocity component increases sharply within approximately 5–10 ms and subsequently decays over a timescale of about 100 ms toward its initial value. Notably, the discharge remains continuously active after ignition, even though its impact on the flow diminishes following this transient phase. The vertical velocity then returns to a steady-state level that is again determined solely by the imposed gas flow rate.

All time-averaged analyses presented in this work therefore exclude the initial ignition phase and only consider the steady-state period, ensuring that transient effects are not included in the averaging.

These results demonstrate that the discharge does influence the flow dynamics. The particularly strong impact of the first pulse is consistent with previous studies, which have shown that the absence of a memory effect in the initial pulse leads to more intense discharge events with higher electric fields and enhanced emission [11, 15]. However, the induced vertical velocities are approximately two orders of magnitude smaller than the dominant horizontal flow velocity. Moreover, the rapid re-establishment of equilibrium indicates that the plasma-induced effect is significant only during the ignition phase and does not lead to a sustained modification of the overall flow field.

At first glance, this result appears surprising, as significantly stronger influences of the plasma on the flow field have been observed and modeled in comparable DBD reactors [19,

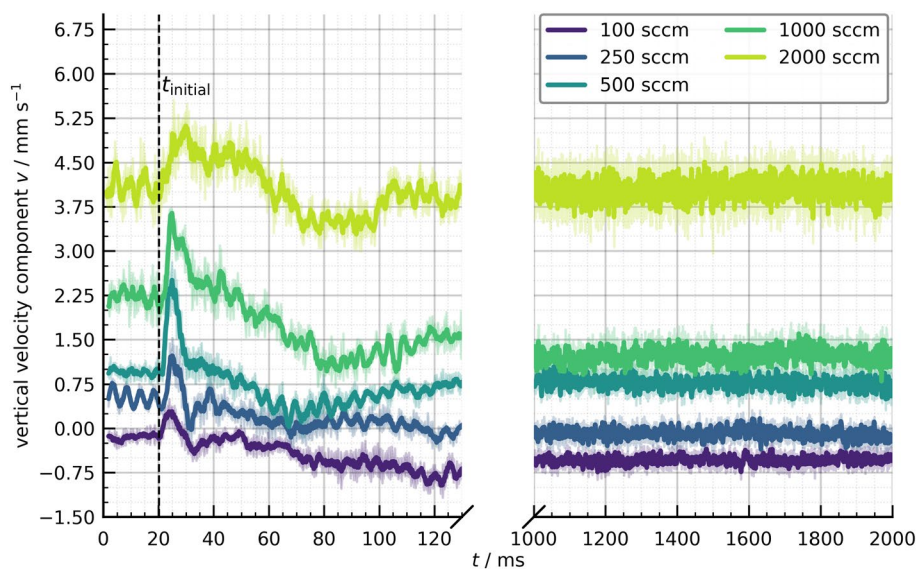


Fig. 6 Temporal evolution of the vertical velocity component v spatially averaged in the gas domain directly above the cavities as a function of operating time and external gas flow rate. t_{initial} represents the point in time of the first discharge ignition. The displayed values are spatially averaged over a rectangular region extending across the full width of the measurement area, approximately 1 mm in height and located about 0.25 mm above the surface (to avoid measuring in an area where aerosol is vanishing due to thermal convection)

[31, 32]. The main difference lies in the microcavity structure of the reactor used here. In the MCPA configuration, the entire discharge is confined within the microcavities, where high electric fields and elevated electron energies are present [9, 10, 13, 15]. However, at atmospheric pressure, the discharge extends only a few micrometers beyond the cavity opening (with a diameter of 200 μm) [8, 33], resulting in a very small interaction volume with the bulk gas flow. In contrast, conventional surface DBD configurations allow the plasma to interact directly with the surrounding gas along exposed electrode surfaces. In such systems, electrohydrodynamic (EHD) forces may extend over centimeter-scale distances along the surface that is directly exposed to the gas flow [32]. In this context, the effective gas treatment volume refers to the region in which the gas flow is directly exposed to and interacts with the plasma. Consequently, this volume is not only significantly larger than in microcavity-based systems, but the interaction can also be stronger, as the plasma volume is directly exposed to the gas flow.

In addition to the EHD force, a purely thermal contribution can be considered as a possible explanation for the observed transient flow. As a simplified upper-bound estimate, assuming that the full applied discharge power of 3.5 W is deposited into the reactor volume of 4 mL, an energy of 0.035 J is introduced within the first 10 ms. Under ideal-gas assumptions for helium and neglecting all losses and plasma-specific sinks, this would correspond to a moderate overall temperature increase of roughly 10–20 K if the energy were homogeneously distributed. In reality, however, the energy deposition is initially strongly localized within the microcavities. The total cavity volume of about 3400 cavities (200 μm diameter, 50 μm depth) corresponds to approximately 5.3 μL , which is very small compared to the

total reactor volume. In this confined region, the same energy input can, in the idealized limit, lead to very high transient temperatures on the order of several thousand Kelvin.

This strong local heating implies a rapid thermal expansion and the formation of a short-lived pressure increase, which can generate a transient pressure pulse. Such a pressure-driven expansion provides a plausible explanation for the observed brief vertical flow following plasma ignition. However, this estimate represents an upper bound, since a significant fraction of the energy is consumed by plasma generation, excitation, ionization, and losses to surrounding dielectric and electrode surfaces.

Experimentally, the transient flow is observed only within the first 10 ms after ignition, followed by a transition to a quasi-steady state. This can be attributed to the rapid establishment of thermal and convective equilibration with the surrounding solid structures and the main flow, which limits further pressure build-up.

Overall, this provides a consistent qualitative picture in which a short-lived thermally induced pressure pulse, superimposed on EHD effects, contributes to the observed transient vertical flow, although the full plasma–flow–thermal coupling is not yet fully resolved.

Influence of Excitation Frequency on Flow Dynamics

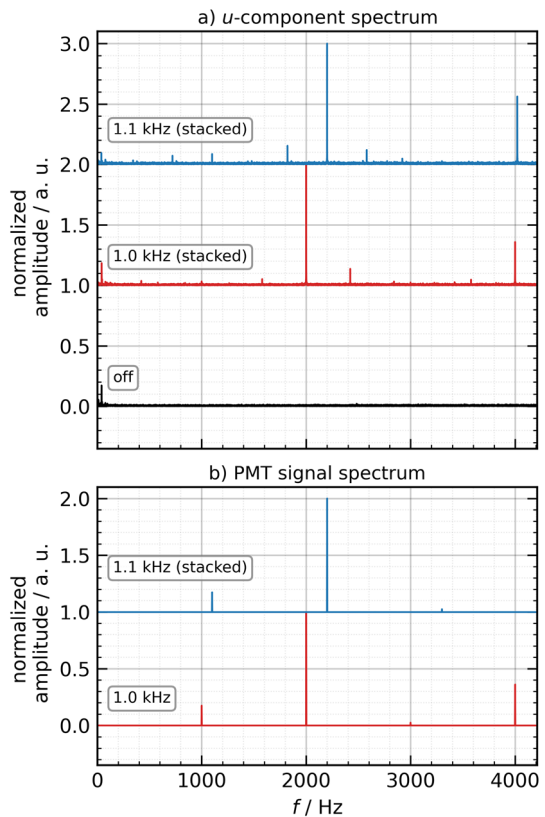
The results show that the discharge can influence the gas dynamics, although the effect is relatively small. To further investigate the influence of the discharge on the gas flow, the vertical velocity is analyzed over a longer time period, as already described above. From Fig. 6, it became clear that after the initial ignition, a steady state is established, in which the vertical velocity fluctuates around a constant mean value. In the following, these fluctuations are examined in more detail by means of a frequency analysis.

For this purpose, the vertical velocity is recorded again over a period of 2 s, but this time exclusively during the steady-state regime. In order to make the influence of the discharge visible, two different excitation frequencies, 1.0 kHz and 1.1 kHz, are used. The excitation frequency was deliberately reduced compared to the standard operating frequency of 15 kHz in order to obtain a larger number of measurement points within a single period.

The results are shown in Fig. 7a. For the plasma-off case, only small frequency components below 50 Hz are visible, which may, for example, originate from the PIV system itself. When the discharge is operated at 1.0 kHz, several peaks appear in the frequency spectrum. The most dominant peak is located exactly at 2 kHz, corresponding to twice the excitation frequency. This observation is consistent with the behavior of the discharge, since ignition occurs twice per period: once during the positive half-cycle and once during the negative half-cycle. Additional peaks can be seen at the excitation frequency as well as at other harmonics. When the excitation frequency is increased to 1.1 kHz, a similar spectrum is observed. However, the dominant peak shifts to 2.2 kHz, which again corresponds to twice the excitation frequency.

To facilitate the interpretation of the spectra, the plasma emission was recorded using a photomultiplier tube, and a frequency analysis was performed as well (see Fig. 7b). The measurement time was set to 2 s, consistent with the PIV measurements. The results clearly show the same dominant frequencies observed previously, thereby confirming the influence of the plasma.

Fig. 7 Normalized frequency spectra of the space-averaged horizontal velocity component u near the array's surface (a) and temporal plasma emission measured with a photomultiplier tube (PMT) (b). Two discharge pulse repetition frequencies were applied for both Fourier analyses: 1.0 kHz and 1.1 kHz



The remaining, less dominant peaks also shift with the excitation frequency and are not present in the plasma-off case. Unfortunately, these peaks are not visible in the emission measurement and could not be identified in more detail.

Conclusion

In this study, the flow dynamics within a microcavity plasma array (MCPA) reactor were investigated using particle image velocimetry (PIV), allowing a detailed analysis of the influence of the plasma discharge on the gas flow. The measurements demonstrate that the discharge has a measurable but overall limited impact on the flow field. The primary effect arises from the increased gas temperature within the discharge channel, which results in a locally elevated flow velocity.

A comparison of spatially resolved measurements under plasma-on and plasma-off conditions shows that deviations from the otherwise laminar velocity profile occur only in the immediate vicinity of the electrode. These perturbations are confined to a region within approximately 0.5 mm from the electrode surface, while the flow further away remains largely unaffected.

Time-resolved measurements further reveal that the ignition of the discharge temporarily influences the flow dynamics. However, the system rapidly returns to a steady state, with

equilibrium conditions being re-established within approximately 100 ms after ignition. Additionally, the excitation frequency of the discharge is reflected in the velocity signal, indicating that the periodic plasma operation also imprints a detectable signature on the flow. These transient and periodic effects warrant further investigation, particularly with regard to their potential impact on conversion efficiency. For instance, burst-mode excitation could be employed to enhance ignition-related effects and systematically assess their influence.

Overall, the influence of the discharge on the flow field, beyond the thermal effects, is minimal and spatially confined. The general flow behavior can therefore be well described by a classical parabolic Poiseuille velocity profile, largely independent of plasma operation or the applied gas flow rate. These findings are in excellent agreement with previous studies on MCPA reactors. In particular, the transport of reactive species and the residence time within the reactor have consistently been successfully approximated using a laminar flow model, and no studies to date have reported additional transport mechanisms such as plasma-induced vortices.

The present results further emphasize the suitability of the MCPA reactor as a model system for plasma-catalytic research. Although laminar flow limits plasma-induced mixing that could enhance conversion, the reactor design deliberately prioritizes well-defined and controllable transport conditions. This makes it a robust platform for fundamental investigations, particularly of plasma–surface interactions. In this context, the predictable and well-characterized transport of reactive species represents a key advantage.

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Data Availability The data generated and analyzed during the current study are available at the following URL: <https://rdpcidat.rub.de/node/1357>.

Declarations

Competing interests The authors declare no competing interests.

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References

1. Belkessa N, Assadi AA, Bouzaza A, Nguyen-Tri P, Amrane A, Khezami L (2024) A review of non-thermal plasma -catalysis: The mutual influence and sources of synergetic effect for boosting volatile organic compounds removal. *Environ Res* 257:119333. <https://doi.org/10.1016/j.envres.2024.119333>
2. Snoeckx R, Bogaerts A (2017) Plasma technology – a novel solution for CO₂ conversion? *Chem Soc Rev* 46(19):5805–5863. <https://doi.org/10.1039/C6CS00066E>
3. Baskaran D, Dhamodharan D, Behera US, Byun H-S (2024) A comprehensive review and perspective research in technology integration for the treatment of gaseous volatile organic compounds. *Environ Res* 251:118472. <https://doi.org/10.1016/j.envres.2024.118472>
4. Bogaerts A, Tu X, Whitehead JC, Centi G, Lefferts L, Guaitella O, Azzolina-Jury F, Kim H-H, Murphy AB, Schneider WF, Nozaki T, Hicks JC, Rousseau A, Thevenet F, Khacef A, Carreon M (2020) The 2020 plasma catalysis roadmap. *J Phys D Appl Phys* 53(44):443001. <https://doi.org/10.1088/1361-6463/ab9048>
5. Hinde P, Demidyuk V, Gkelios A, Tipton C (2020) Plasma Catalysis: A Review of the Interdisciplinary Challenges Faced: Realising the potential of plasma catalysis on a commercial scale. *Johnson Matthey Technol Rev* 64(2):138–147. <https://doi.org/10.1595/205651320X15759961130711>
6. Lefferts L (2024) Leveraging Expertise in Thermal Catalysis to Understand Plasma Catalysis. *Angew Chem Int Ed* 63(10):202305322. <https://doi.org/10.1002/anie.202305322>
7. De Meyer R, Gorbanev Y, Ciocarlan R-G, Cool P, Bals S, Bogaerts A (2024) Importance of plasma discharge characteristics in plasma catalysis: Dry reforming of methane vs. ammonia synthesis. *Chem Eng J* 488:150838. <https://doi.org/10.1016/j.cej.2024.150838>
8. Dzikowski S, Michaud R, Böttner H, Dussart R, Böke M, Schulz-von der Gathen V (2020) Modular constructed metal-grid arrays-an alternative to silicon-based microplasma devices for catalytic applications. *Plasma Sources Sci Technol* 29(3):035028. <https://doi.org/10.1088/1361-6595/ab71f6>
9. Dzikowski S, Steuer D, Iséni S, Golda J, Böke M, Schulz-von der Gathen V (2022) Electric field strengths within a micro cavity plasma array measured by Stark shift and splitting of a helium line pair. *Plasma Sources Sci Technol* 31(6):065014. <https://doi.org/10.1088/1361-6595/ac7820>
10. van Impel H, Steuer D, Labenski R, Schulz-von der Gathen V, Böke M, Golda J (2024) Electric field components within a micro-scaled DBD measured by Stark shifting and splitting of helium lines. *Plasma Sources Sci Technol* 33(10):105008. <https://doi.org/10.1088/1361-6595/ad7d34>
11. Labenski R, Steuer D, van Impel H, Böke M, Schulz-von der Gathen V, Golda J (2024) Novel methods for determination and manipulation of surface charges performed on an atmospheric DBD microplasma. *Plasma Sources Sci Technol* 33(10):105022. <https://doi.org/10.1088/1361-6595/ad802d>
12. Steuer D, van Impel H, Labenski R, Schulz-von der Gathen V, Böke M, Golda J (2025) Mode transition in a helium barrier discharge with oxygen admixtures: insights into a micro cavity plasma array reactor. *J Phys D Appl Phys* 58(8):085211. <https://doi.org/10.1088/1361-6463/ad9ebe>
13. Steuer D, van Impel H, Schulz-von der Gathen V, Böke M, Golda J (2023) Spatially and temporally resolved atomic oxygen densities in a micro cavity plasma array. *Plasma Sources Sci Technol* 32(2):025013. <https://doi.org/10.1088/1361-6595/acb9b9>
14. Steuer D, Bentz BZ, Youngman K, van Impel H, Böke M, Schulz-von Der Gathen V, Golda J (2024) Measurement of atomic oxygen densities using TALIF on a dielectric barrier discharge: insights into the volume above a micro cavity plasma array. *Plasma Sources Sci Technol* 33(12):125007 <https://doi.org/10.1088/1361-6595/ad97da>
15. van Impel H, Steuer D, Schulz-von der Gathen V, Böke M, Golda J (2025) Analyzing atomic oxygen product evolution in micro cavity plasma arrays by a combination of a multi-PMT OES setup and a 0D chemical model. *Plasma Sources Sci Technol* 34(7):075012. <https://doi.org/10.1088/1361-6595/adeec0>
16. van Impel H, Krettek O, Steuer D, Schulz-von der Gathen V, Böke M, Golda J (2026) 3d co-talif distribution above a micro cavity discharge: A systematic approach for plasma catalysis. submitted to PSST <https://doi.org/10.48550/arXiv.2605.18217>
17. Robinson M (1962) A history of the electric wind. *Am J Phys* 30(5):366–372. <https://doi.org/10.1119/1.1942021>
18. Roth JR (2003) Aerodynamic flow acceleration using piezoelectric and peristaltic electrohydrodynamic effects of a One Atmosphere Uniform Glow Discharge Plasma. *Phys Plasmas* 10(5):2117–2126. <https://doi.org/10.1063/1.1564823>
19. Böddecke A, Passmann M, Segura ANT, Bodnar A, Awakowicz F, Oppotsch T, Muhler M, Awakowicz P, Gibson AR, Korolov I, Mussenbrock T (2024) The role of flow field dynamics in enhancing volatile organic compound conversion in a surface dielectric barrier discharge system. *J Phys D Appl Phys* 58(2):025208. <https://doi.org/10.1088/1361-6463/ad8454>

20. Robert E, Sarron V, Darny T, Riés D, Dozias S, Fontane J, Joly L, Pouvesle J-M (2014) Rare gas flow structuration in plasma jet experiments. *Plasma Sources Sci Technol* 23(1):012003. <https://doi.org/10.1088/0963-0252/23/1/012003>
21. Kawasaki T, Kusumegi S, Kudo A, Sakanoshita T, Tsurumaru T, Sato A, Uchida G, Koga K, Shiratani M (2016) Effects of irradiation distance on supply of reactive oxygen species to the bottom of a petri dish filled with liquid by an atmospheric o₂/he plasma jet. *J Appl Phys* 119(17):173301. <https://doi.org/10.1063/1.4948430>
22. Stancampiano A, Bocanegra PE, Dozias S, Pouvesle J-M, Robert E (2021) Evidence, origin and impact of liquid flows in plasma medicine in vitro treatments with APPJs. *Plasma Sources Sci Technol* 30(1):015002. <https://doi.org/10.1088/1361-6595/abcc50>
23. Raffel M, Willert CE, Scarano F, Kähler CJ, Wereley ST, Kompenhans J (2018) *Particle Image Velocimetry: A Practical Guide*. Springer, Cham. <https://doi.org/10.1007/978-3-319-68852-7>
24. Smolik J, Žďimal V (1993) Condensation of supersaturated vapors. Homogeneous nucleation of bis(2-ethyl-hexyl)sebacate(DEHS). *J Aerosol Sci* 24(5)
25. Skourides C, Nyfantis D, Leyland P, Bosse H, Ott P (2019) Mechanisms of control authority by nano-second pulsed dielectric barrier discharge actuators on flow separation. *Appl Sci Basel* 9(15):2989
26. Scharnowski S, Kähler CJ (2020) Particle image velocimetry - classical operating rules from today's perspective. *Opt Lasers Eng* 135:106185. <https://doi.org/10.1016/j.optlaseng.2020.106185>
27. Westerweel J, Scarano F (2005) Universal outlier detection for piv data. *Exp Fluids* 39:1096–1100. <https://doi.org/10.1007/s00348-005-0016-6>
28. Thielicke W, Stamhuis E (2014) PIVlab-towards user-friendly, affordable and accurate digital particle image velocimetry in MATLAB. *J Open Res Softw* 2(1):30. <https://doi.org/10.5334/jors.bl>
29. Thielicke W, Sonntag R (2021) Particle image velocimetry for MATLAB: accuracy and enhanced algorithms in PIVlab. *J Open Res Softw*. <https://doi.org/10.5334/jors.334>
30. Kundu PK, Cohen IM, Dowling DR (2012) *Fluid Mechanics*, 5th, ed edn. Elsevier Academic Press, Amsterdam Heidelberg
31. Mohsenimehr S, Von Keudell A (2023) Surface Dielectric Barrier Discharge (sDBD) for flow control in plasma conversion. *Plasma Chem Plasma Process* 43(6):1633–1649. <https://doi.org/10.1007/s11090-023-10405-z>
32. Mohsenimehr S, Wilczek S, Mussenbrock T, Keudell AV (2025) Plasma and flow simulation of the ion wind in a surface barrier discharge used for gas conversion benchmarked by Schlieren imaging. *Plasma Chem Plasma Process* 45(1):85–112. <https://doi.org/10.1007/s11090-024-10533-0>
33. Kreuznacht S, Böke M, Schulz-von der Gathen V (2021) Intracavity dynamics in a microplasma channel by sideon imaging. *Plasma Sources Sci Technol* 30(1):015014. <https://doi.org/10.1088/1361-6595/abdb1e>

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